

## No. of Functional Features Included in the Solution

|                      |              |
|----------------------|--------------|
| <b>Date</b>          | 28 June 2026 |
| <b>Team ID</b>       |              |
| <b>Project Name</b>  | Smart Lender |
| <b>Maximum Marks</b> | 5 Marks      |

### Functional Features Overview:

The Smart Lender system provides loan eligibility prediction using machine learning and a simple web interface.

It includes data preprocessing, prediction, and result display features. All core features are implemented and tested successfully, making the system reliable and user-friendly.

| S.No | Feature Name                | Feature Description                                       | Module / Component   | Status (Done / In Progress / Pending) | Marks Contribution |
|------|-----------------------------|-----------------------------------------------------------|----------------------|---------------------------------------|--------------------|
| 1    | User Input Form             | Collects applicant details for loan prediction            | Frontend             | Done                                  | High               |
| 2    | Data Preprocessing          | Cleans and prepares data for model training               | Preprocessing Module | Done                                  | High               |
| 3    | Exploratory Data Analysis   | Performs univariate, bivariate, and multivariate analysis | EDA Module           | Done                                  | Medium             |
| 4    | Machine Learning Model      | Trains and evaluates the loan prediction model            | ML Module            | Done                                  | High               |
| 5    | Loan Eligibility Prediction | Predicts whether a loan is approved or rejected           | Prediction Module    | Done                                  | High               |
| 6    | Confidence Score Display    | Displays prediction confidence percentage                 | Backend              | Done                                  | Medium             |
| 7    | Responsive User Interface   | Provides a user-friendly web interface                    | Frontend             | Done                                  | Medium             |
| 8    | Error Handling              | Handles invalid inputs and exceptions                     | Backend              | Done                                  | Medium             |

**Feature Summary:**

| <b>Metric</b>                      | <b>Count / Value</b> |
|------------------------------------|----------------------|
| Total Features Planned             | 8                    |
| Total Features Implemented         | 8                    |
| Core / Must-Have Features          | 5                    |
| Additional / Nice-to-Have Features | 3                    |
| Features Tested & Verified         | 8                    |

**Feature Category Breakdown:**

| <b>S.No</b> | <b>Category</b>           | <b>Features in Category</b> | <b>Example Features</b>                            |
|-------------|---------------------------|-----------------------------|----------------------------------------------------|
| 1           | User Interface (UI)       | 3                           | Home Page, Input Form, Result Page                 |
| 2           | Backend / Logic           | 3                           | Prediction Logic, Confidence Score, Error Handling |
| 3           | Database / Storage        | 1                           | Model and Scaler Storage (.pkl files)              |
| 4           | API / Integration         | 1                           | Flask Routes and Prediction API                    |
| 5           | Security / Authentication | 0                           | Not Implemented                                    |